

# Hysteresis between Bitcoin, Gold, Oil, and the S&P 500 Index

Evidence from a multi-threshold connectedness approach

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## Introduction

Understanding the interconnectedness of major financial markets has become increasingly critical in the face of recurring global shocks — the COVID-19 pandemic, geopolitical tensions, and abrupt monetary-policy shifts. Investors and policymakers are concerned not only with the *presence* of cross-market spillovers but also with their *intensity* and *persistence*.

A central but often-overlooked property of these spillovers is **hysteresis** — the idea that shocks leave structural footprints on inter-market relationships that do not unwind once the originating disturbance dissipates (Blanchard & Summers, 1986; Baldwin, 1990; Verheyen, 2013). Behavioural-finance models likewise predict long-lasting changes in market linkages following extreme events (Barberis et al., 1998; Baur & Lucey, 2010). Yet most empirical connectedness studies still implicitly assume *symmetric reversion* of spillovers once turbulence eases — a restriction the data may not support.

Building on the Diebold and Yilmaz spillover framework (Diebold & Yilmaz, 2012, 2014) and quantile-connectedness extensions (Chatziantoniou et al., 2021; Gabauer & Stenfors, 2024), this study develops a **multi-threshold connectedness framework** that aggregates recurring large changes into regime-specific TVP-VAR systems and formally tests for hysteresis.

## Research Question

Does cross-asset connectedness in **tail regimes** differ structurally from connectedness in **stable regimes**, and does it fail to revert symmetrically once turbulence eases?

Two testable implications follow:

- Tail-regime connectedness differs significantly from stable-regime connectedness.
- Increasing and decreasing regimes need not behave symmetrically — directional dominance may differ.

## Contribution

The proposed framework differs from existing quantile-based approaches in three respects. First, it **aggregates** cumulative large changes into separate regime-specific series, aligning directly with how investors react to recurring panic episodes rather than to isolated quantile bins. Second, it preserves **pairwise connectedness** measures often dropped under quantile-on-quantile methods for computational tractability. Third, it embeds a **formal hysteresis test** via two-sided *t*-tests of regime-mean connectedness measures, providing second-order inference absent from earlier work.

## Methods

### Data and Sample

Daily returns on **Bitcoin, Gold, WTI Oil, and the S&P 500 Index** from **4 January 2015 to 21 May 2024** are collected from *investing.com*. After aligning trading days across markets, the final sample contains **2,360 common daily observations**. All four series are stationary (ADF), reject normality (Jarque–Bera), are negatively skewed, and exhibit serial dependence consistent with daily-return autocorrelation in finance (Avramov et al., 2006; Lewellen, 2002).

### Three-Regime Decomposition

Following Verheyen (2013), each return series  $y_t$  is partitioned into three regimes using quantile thresholds  $k_2 = Q(30\%)$  and  $k_1 = Q(70\%)$ :

$$\begin{aligned} y_t^+ &= \sum_{i=1}^t \Delta y_i \cdot \mathbb{I}\{\Delta y_i > k_1\} && \text{(increasing)} \\ y_t^\pm &= \sum_{i=1}^t \Delta y_i \cdot \mathbb{I}\{k_2 \leq \Delta y_i \leq k_1\} && \text{(stable)} \\ y_t^- &= \sum_{i=1}^t \Delta y_i \cdot \mathbb{I}\{\Delta y_i < k_2\} && \text{(decreasing)} \end{aligned}$$

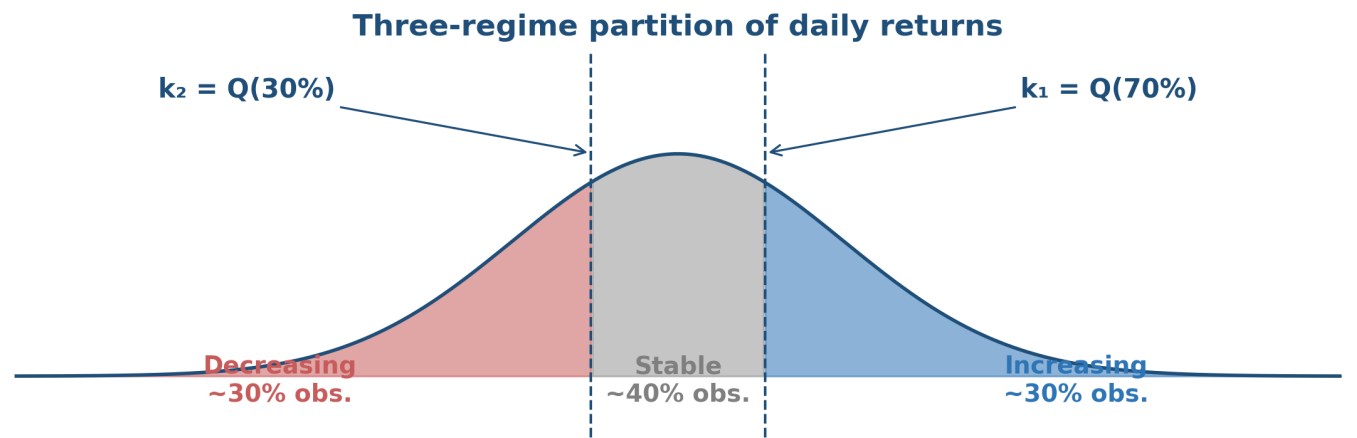


Figure 1. Three-regime partition of daily returns. The 30/70 split allocates roughly 30% of observations to each tail regime and 40% to the stable centre, balancing tail capture with estimation stability.

## TVP-VAR Connectedness

A time-varying parameter VAR (Antonakakis et al., 2020) is estimated separately on each regime’s partial-sum series. Generalized forecast-error variance decompositions (Pesaran & Shin, 1998) are then used to construct the standard connectedness measures: the total connectedness index (TCI), gross transmission (TO), gross reception (FROM), the net position (NET), and net pairwise directional connectedness (NPDC). The TVP-VAR removes the need for an arbitrary rolling-window choice and is robust to outliers, particularly important when estimating dynamics on tail-regime subsamples.

**Hysteresis test.** For every connectedness measure, regime-mean values are compared via two-sided *t*-tests across the increasing-versus-stable and decreasing-versus-stable contrasts. If both contrasts are significant at the 5% level, hysteresis is identified for that measure.

## Results

### Aggregate Connectedness

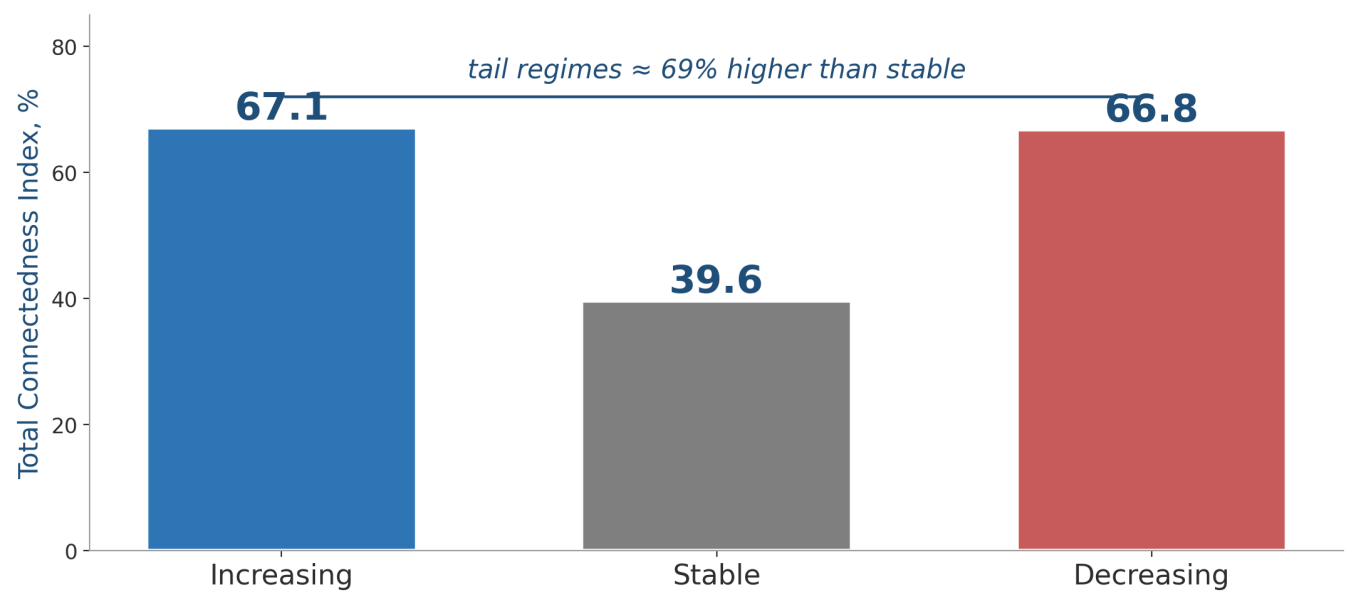


Figure 2. Total Connectedness Index by regime. Tail-regime connectedness is roughly 69% higher than stable-regime connectedness, and the two tail regimes are themselves of similar magnitude.

System-wide spillovers are markedly higher under turbulence: TCI reaches **67.1%** in the increasing regime and **66.8%** in the decreasing regime, against only **39.6%** in the stable regime. The stable-regime trough between 10 December 2018 and 28 August 2019 (mean TCI  $\approx 12$ , minimum 6.6) suggests diversification benefits that are essentially regime-conditional.

### Directional Roles

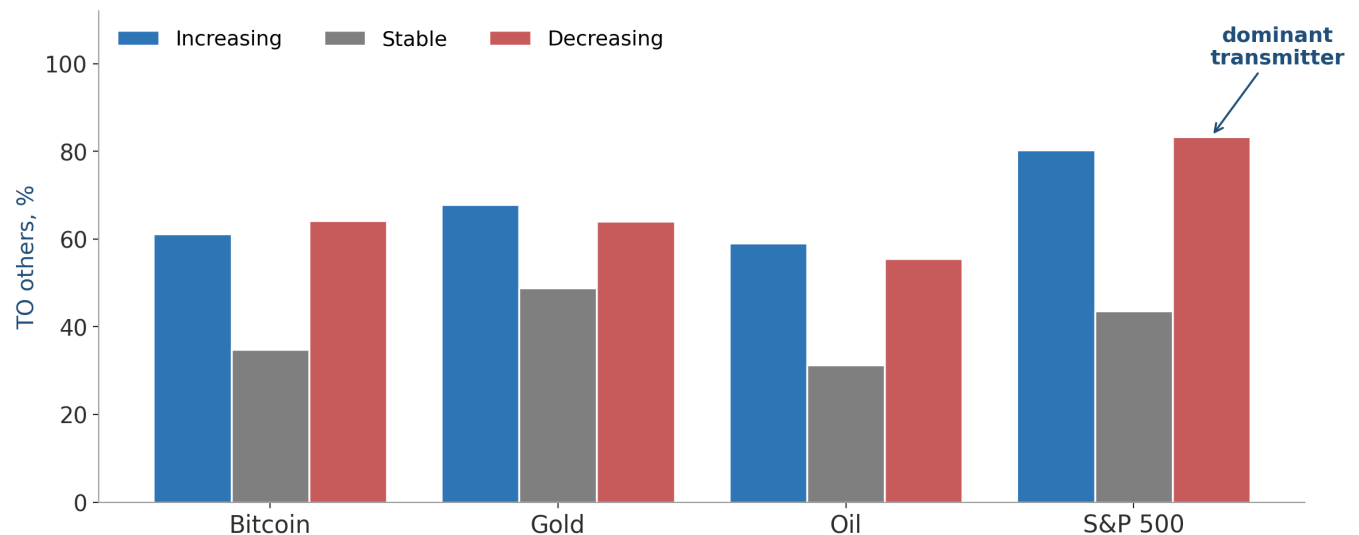


Figure 3. Gross transmission TO others, by regime and asset. The S&P 500 is the dominant transmitter in every regime and the only asset with positive NET in both tail regimes (+15.6 and +20.0).

The **S&P 500** consistently emerges as the systemic centre, transmitting more shocks than it absorbs across every regime — reinforcing the role of US equities as a primary conduit of global financial contagion (Kang et al., 2019; Yousaf et al., 2023).

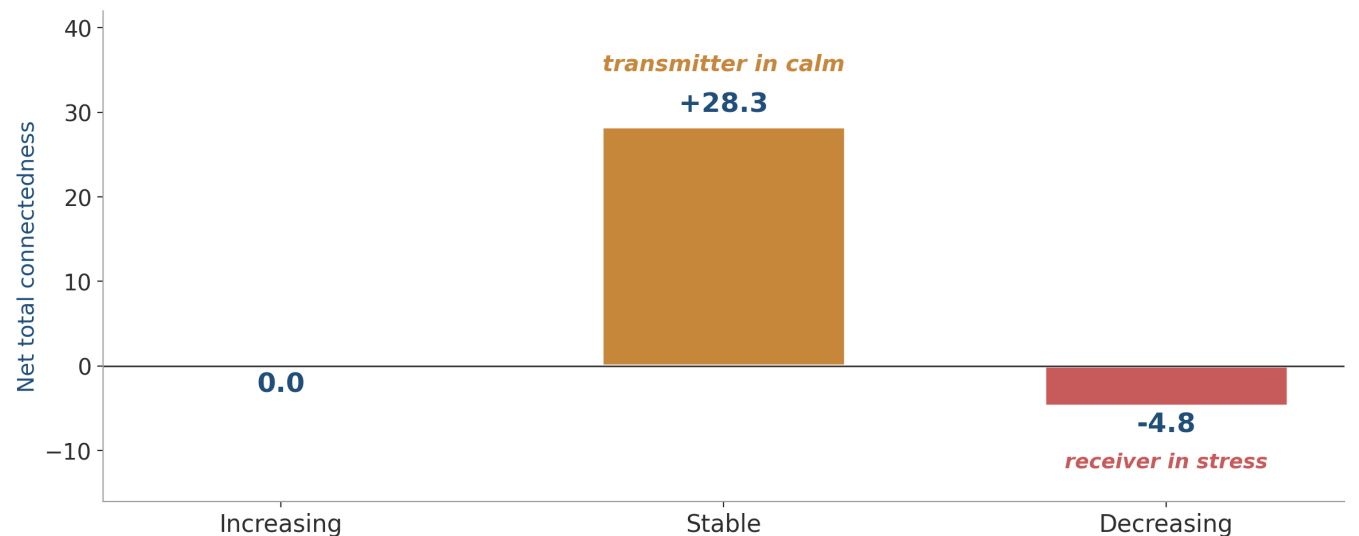


Figure 4. Gold’s net connectedness reverses sign across regimes: +28.3 in calm but –4.8 in stress. The safe-haven role is therefore conditional, not unconditional.

The most striking finding concerns **Gold**, whose role reverses by regime. In calm conditions Gold is a net transmitter (NET = +28.3), likely reflecting its influence on hedging-portfolio rebalancing flows. Under stress it switches to net receiver (NET  $\in \{-0.0, -4.8\}$ ), absorbing flight-to-safety inflows. This qualifies the popular view of Gold as an *unconditional* safe haven (Będowska-Sójka & Kliber, 2021): the property is **regime-contingent**.

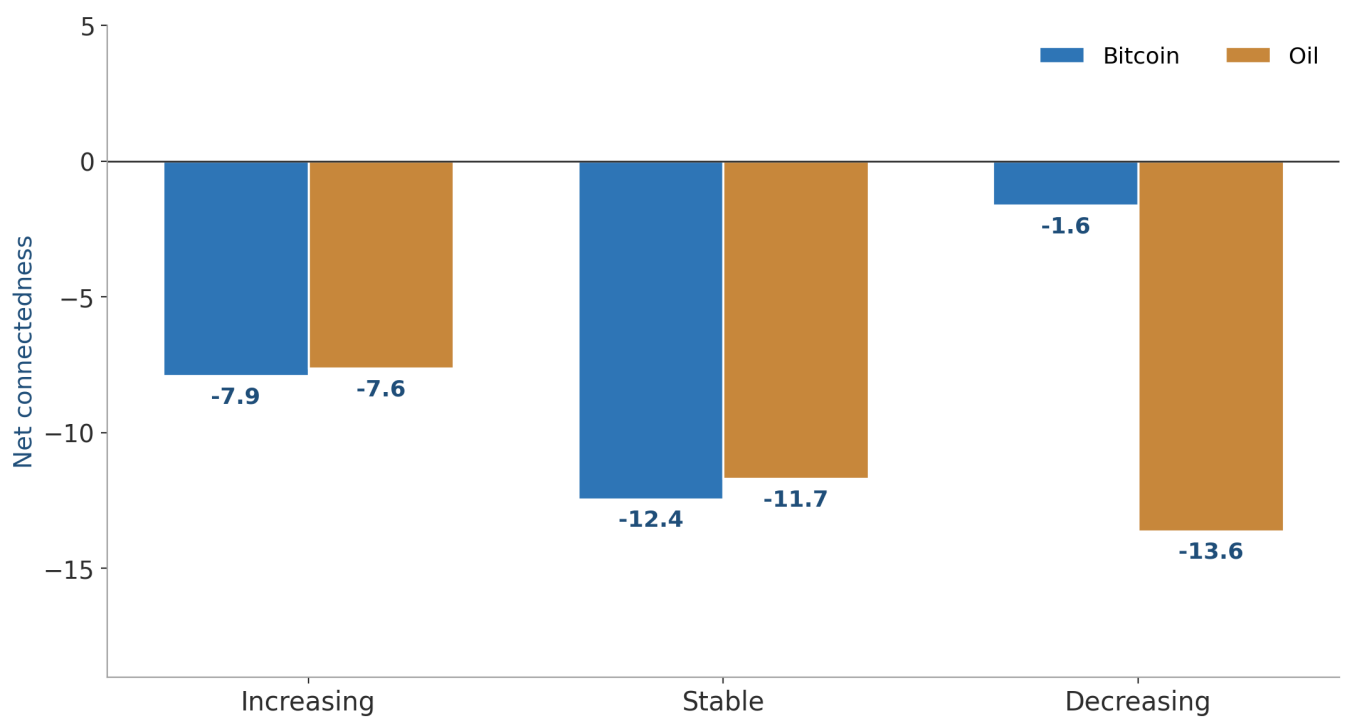


Figure 5. Bitcoin and Oil are persistent net receivers across all three regimes. Bitcoin’s deficit narrows in stress, while Oil’s deepens.

**Bitcoin** remains a net receiver across all regimes (NET  $\in \{-12.5, -7.9, -1.6\}$ ), yet its gross spillover intensity *rises* markedly in stress — a signature of deepening integration into mainstream finance (Guo, 2024). **Oil** is the most vulnerable asset of the four, with NET deteriorating to –13.6 in the decreasing regime, contrary to its conventional framing as a shock source for the wider economy.

## Hysteresis Tests

| Measure                        | Inc. vs Stab. | Dec. vs Stab. |
|--------------------------------|---------------|---------------|
| TCI (total connectedness)      | 78.4***       | 79.0***       |
| Gold — NET                     | –17.4***      | –21.5***      |
| S&P 500 — NET                  | 18.3***       | 20.7***       |
| Oil — FROM others              | 43.7***       | 53.0***       |
| S&P 500 $\leftrightarrow$ Gold | 80.4***       | 83.4***       |

Two-sided *t*-tests of regime-mean connectedness. \*\*\* significant at 1%. Hysteresis is confirmed for every measure shown.

## Robustness

Re-estimating the system at alternative quantile cutoffs (25/75 and 20/80) preserves the qualitative result: tail-regime TCI remains in the 66–67% range while stable-regime TCI falls further as the inner corridor narrows. The gap between tail and stable regimes *widens* under tighter cutoffs, strengthening the hysteresis inference rather than weakening it.

## Conclusions

- Hysteresis is confirmed.** Tail-regime connectedness differs significantly from stable-regime connectedness, and does not revert symmetrically when conditions calm.
- The S&P 500 is the systemic centre**, acting as the dominant net transmitter across every regime.
- Gold is conditional, not unconditional** — a net transmitter in calm but a net receiver in stress, qualifying the popular safe-haven view.
- Bitcoin integrates, Oil absorbs.** Bitcoin’s spillovers intensify under stress, signalling deeper financial integration; Oil is the most vulnerable asset in every regime.

Investment practice should treat diversification as *regime-dependent*, stress-test hedging allocations that lean on Gold or Bitcoin for downside protection, and maintain elevated hedge ratios for some period after a stress event since connectedness does not unwind mechanically once volatility eases.

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